

Zelta Premium — Telegram Mini App + Bot

Production-Ready Lead Funnel System

Premium furniture brand lead capture system built as a Telegram Mini App with an integrated Telegram Bot and Admin Panel.

Architecture Overview

Component	Technology	Purpose
Mini App Frontend	Vanilla HTML/CSS/JS	6-screen bilingual lead funnel (Uzbek + Russian)
Backend API	Node.js + Express	REST API for leads, file uploads, admin
Database	SQLite (sql.js)	Lead storage, status tracking, audit log
File Storage	Local filesystem	Uploaded room photos stored in <code>/public/uploads/</code>
Telegram Bot	node-telegram-bot-api	<code>/start</code> entry, Mini App launcher, admin notifications
Admin Panel	HTML/CSS/JS	Lead management, status updates, CSV export

Project Structure

```
zelta-premium/
├─ server.js           # Express server + API routes
├─ bot.js              # Telegram Bot (separate process)
├─ package.json
├─ public/
│   ├─ index.html      # Mini App – main funnel UI
│   ├─ app.js          # Mini App – frontend logic
│   ├─ uploads/        # Uploaded photos (auto-created)
│   └─ images/         # Static assets
├─ admin/
│   └─ index.html      # Admin Panel
├─ data/
│   └─ zelta.db        # SQLite database (auto-created)
└─ src/
    └─ locales/
        └─ translations.js # Full Uzbek + Russian translations
```

Funnel Flow (6 Screens)

Screen	Purpose	Key Elements
1. Landing	Brand intro + trust	Language switcher, logo, call card, showroom, Instagram/Telegram links, privacy line
2. Offer	Explain limited offer	Counter (36th client), 3-step process explanation
3. Photos	Collect 3 room photos	Upload grid, tips, preview, replace functionality
4. Phone	Capture phone number	+998 prefix, formatting, privacy reassurance
5. Qualification	Optional furniture/room type	Low-friction selection grid, skip option
6. Success	Confirmation	20-min SLA promise, portfolio + call buttons

Setup & Deployment

Prerequisites

- Node.js 18+ installed
- A Telegram Bot token (from [@BotFather](#))
- A public HTTPS URL for the Mini App (required by Telegram)

1. Install Dependencies

```
cd zelta-premium
npm install
```

2. Configure Environment Variables

```
# Required for Telegram Bot
export TELEGRAM_BOT_TOKEN="your_bot_token_here"

# Required for Mini App URL (must be HTTPS)
export MINI_APP_URL="https://your-domain.com"

# Optional: Admin notifications
export ADMIN_CHAT_ID="your_admin_chat_id"

# Optional: Custom port (default: 3000)
export PORT=3000
```

3. Start the Server

```
# Start the web server (Mini App + API + Admin)
npm start

# In a separate terminal, start the bot
npm run bot
```

4. Configure Telegram Bot

1. Open [@BotFather](#) in Telegram
2. Send `/mybots` → select your bot
3. Go to **Bot Settings** → **Menu Button**
4. Set the menu button URL to your Mini App URL (e.g., `https://your-domain.com`)
5. Go to **Bot Settings** → **Configure Mini App**
6. Set the Mini App URL

5. Deploy to Production

Option A: VPS / Cloud Server

```
# Install PM2 for process management
npm install -g pm2

# Start server
pm2 start server.js --name zelta-server

# Start bot
pm2 start bot.js --name zelta-bot

# Save PM2 config
pm2 save
pm2 startup
```

Option B: Vercel (Frontend) + Railway/Render (Backend)

The project can be split:

- Deploy `public/` to Vercel as static site
- Deploy `server.js` to Railway/Render as Node.js app
- Point API calls to the backend URL

Option C: Docker

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install --production
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]
```

API Reference

Submit Lead

```
POST /api/leads
Content-Type: multipart/form-data

Fields:
- lead_data: JSON string with lead information
- photo_1: Image file
- photo_2: Image file
- photo_3: Image file
```

Get All Leads

```
GET /api/leads?status=new&limit=50&offset=0&search=phone
```

Get Single Lead

```
GET /api/leads/:lead_id
```

Update Lead

```
PATCH /api/leads/:lead_id
```

```
Content-Type: application/json
```

```
Body: { "status": "contacted", "notes": "Called client", "manager_assigned":  
"John" }
```

Export CSV

```
GET /api/export/csv
```

Dashboard Stats

```
GET /api/stats
```

Lead Statuses

Status	Meaning
new	Just submitted, awaiting first contact
waiting	Waiting for client response
contacted	First contact made
render_progress	3D render being prepared
render_sent	Render delivered to client
qualified	Client qualified for sale
closed	Sale completed
lost	Lead lost / not interested

Database Schema

leads Table

Column	Type	Description
lead_id	TEXT PK	Auto-generated (ZP-XXXXX-XXXXXX)
created_at	TEXT	ISO timestamp
telegram_user_id	TEXT	Telegram user ID
telegram_username	TEXT	Telegram @username
full_name	TEXT	First + last name
selected_language	TEXT	'uz' or 'ru'
photo_1, photo_2, photo_3	TEXT	File paths
phone_number	TEXT	+998XXXXXXXXX format
furniture_interest	TEXT	bedroom/living/dining/other
room_type	TEXT	new/renovation/existing
current_status	TEXT	Lead pipeline status
source	TEXT	Traffic source
session_metadata	TEXT	Full JSON session data
submitted_at	TEXT	Form submission time
manager_assigned	TEXT	Assigned manager name
first_contact_at	TEXT	When first contacted
notes	TEXT	Manager notes
sla_deadline	TEXT	20-min callback deadline

lead_status_log Table

Tracks all status changes for audit trail.

Bilingual Support

The app supports full **Uzbek** and **Russian** localization:

- Language switcher on Screen 1 (embedded, not a separate page)
 - All UI text, labels, buttons, tips, and messages are translated
 - Russian translations are polished business-friendly text (not literal translations)
 - Selected language is saved in session and submitted with lead data
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Key Features

- **Premium dark UI:** Black/graphite background with muted gold accents
 - **Mobile-first:** Optimized for Telegram Mini App viewport
 - **Click actions:** Phone call, Instagram link, Telegram deep link with prefilled message
 - **Photo upload:** 3 required photos with preview and replace
 - **Phone validation:** Uzbekistan format (+998) with auto-formatting
 - **20-minute SLA:** Tracked in database with admin dashboard indicator
 - **Admin notifications:** Telegram message to admin on new lead
 - **CSV export:** One-click download of all leads
 - **Status pipeline:** Full lead lifecycle tracking
 - **Auto-refresh:** Admin panel refreshes every 30 seconds
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Access Points

URL	Purpose
<code>http://localhost:3000</code>	Mini App (main funnel)
<code>http://localhost:3000/admin</code>	Admin Panel
<code>http://localhost:3000/api/leads</code>	Leads API
<code>http://localhost:3000/api/stats</code>	Stats API
<code>http://localhost:3000/api/export/csv</code>	CSV Export

Security Notes

- In production, add authentication to `/admin` and `/api/*` routes
 - Use HTTPS (required for Telegram Mini Apps)
 - Consider rate limiting on `/api/leads` POST endpoint
 - Move file uploads to cloud storage (S3, Cloudflare R2) for production
 - Add CORS configuration for production domain
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Future Enhancements

- Dynamic counter (backend-driven, currently static at 36)
- SLA escalation alerts (auto-remind if 20 min exceeded)
- Manager assignment automation
- Webhook integration for CRM systems
- Analytics dashboard with conversion metrics
- A/B testing for funnel copy
- Push notification reminders via Telegram Bot